KESHAV MEMORIAL INSTITUTE OF TECHNOLOGY
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SKILLING-ML/DL
II-II CSE(AI&ML)
AY: 2025-2026

SL No	Week	Slot-1 (3 Hours)	Detailed Subtopics	Slot-2 (3 Hours)	Detailed Subtopics
1	Week 4	Clustering& Dimensionality Reduction (Theory)	K-Means, Hierarchical Clustering, PCA	Computer Vision	Computer Vision and Visual Perception, Applications of Computer Vision, Computer Vision Pipeline Process, Classifier Learning Algorithm.

CONTENTS (Week 4 -Slot 2: Computer Vision**)**

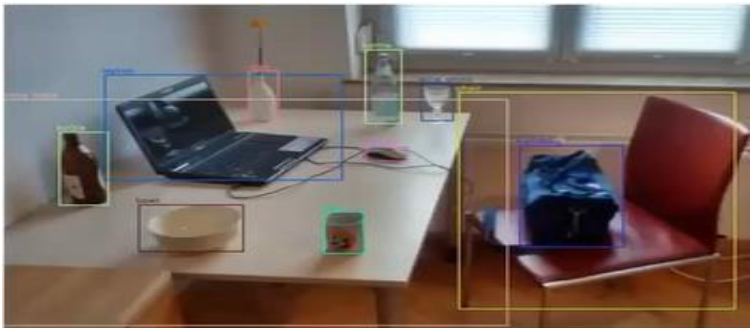
1. Computer Vision and Visual Perception
2. Types of Vision System.
3. Applications of Computer Vision.
4. Computer Vision Pipeline Process.
5. Classifier Learning Algorithm.

1. Computer Vision and Visual Perception

Any AI system can perceive the environment and take actions based on its perceptions.

Computer vision is a field of artificial intelligence (AI) and computer science that focuses on enabling computers to interpret and understand visual information from the world, much like humans do with their eyes and brain.

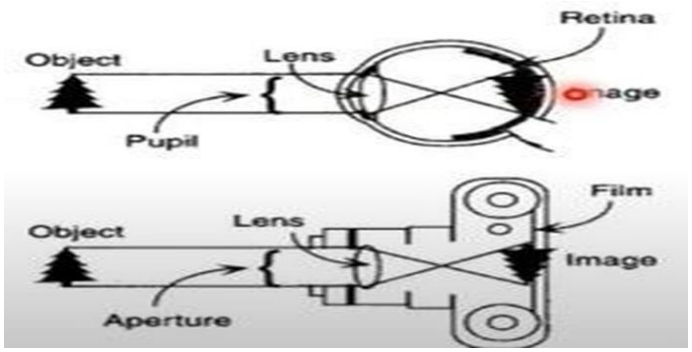
It involves in the *development of algorithms and techniques* that allow machines to extract meaningful insights and make decisions based on images or video data.



Visual perception is the act of observing patterns and objects through sight or visual input. The process of acquiring knowledge about environmental objects and events by extracting information from the light they emit or reflect (Palmer, 2012)

Steps involved in Visual perception:

Analogy behind the capture of images by human and by camera:

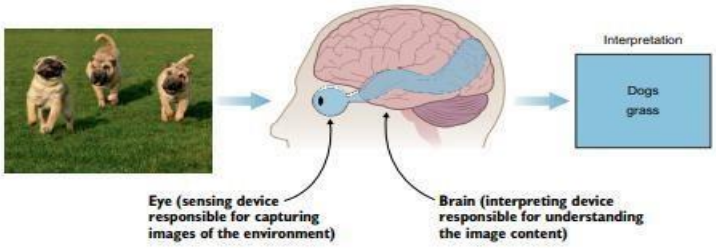


The whole process of visual perception involves not just seeing, but also accurately interpreting what is seen.

2. Types of Vision system:

a. Human Vision Systems:

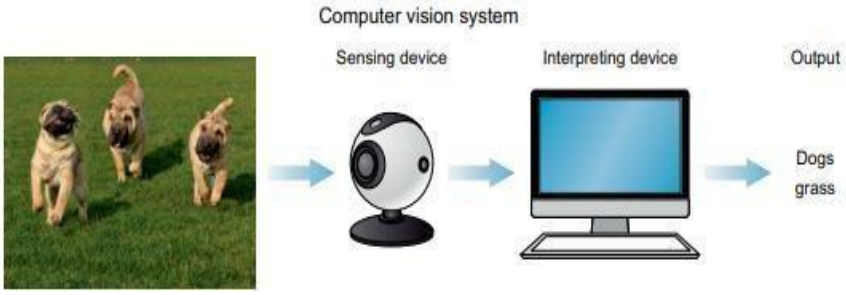
The vision system for humans, animals and most living organisms consists of a sensor or an eye to capture the image and a brain to process & interpret the image. The system outputs a prediction of the image components based on the data extracted from the image.



The human vision system uses the eye and brain to sense and interpret an image.

b. AI Vision Systems:

To replicate human vision, two key components are required: a sensing device that mimics the eye's functionality, and a sophisticated algorithm that replicates the brain's ability to interpret and categorize visual content



The components of the computer vision system are a sensing device and an interpreting device.

Human Vision vs Computer Vision

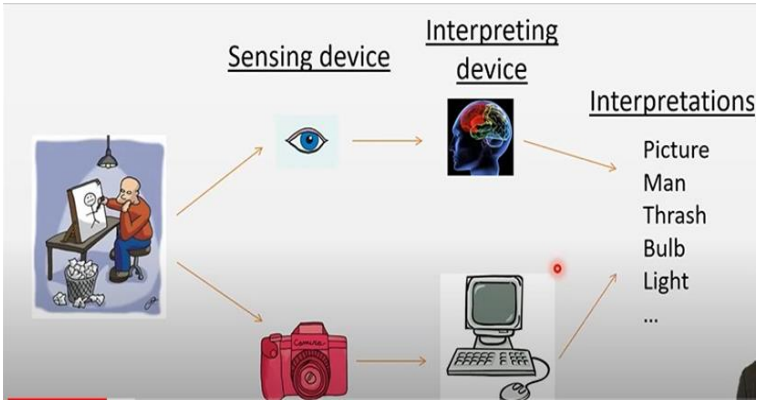
Human Vision System

The human vision system consists of: - Eye (sensor): captures light and images - Brain (interpreter): processes and understands visual data. Humans naturally detect objects, recognize patterns, and understand scenes due to years of learning and experience.

Computer Vision System

A computer vision system mimics human vision using: - Sensing devices (camera, LIDAR, radar, X-ray, CT scan) - Interpreting devices (algorithms, neural networks).

Unlike humans, computers view images as numerical pixel values arranged in matrices.

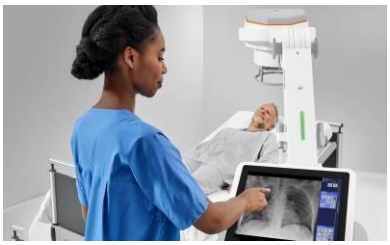


Basic components for Human and Machine Vision Systems: Both human and machine vision systems have two components a sensing device and an interpreting *device*. Each is tailored to fulfil a specific task.

Sensing devices:

The important aspect in selecting a best sensing device to capture the surroundings of a specific environment, whether that is a camera, radar, X-ray, CT scan, Lidar, or a combination of devices to provide the full scene of an environment to fulfill the task at hand.

Camera



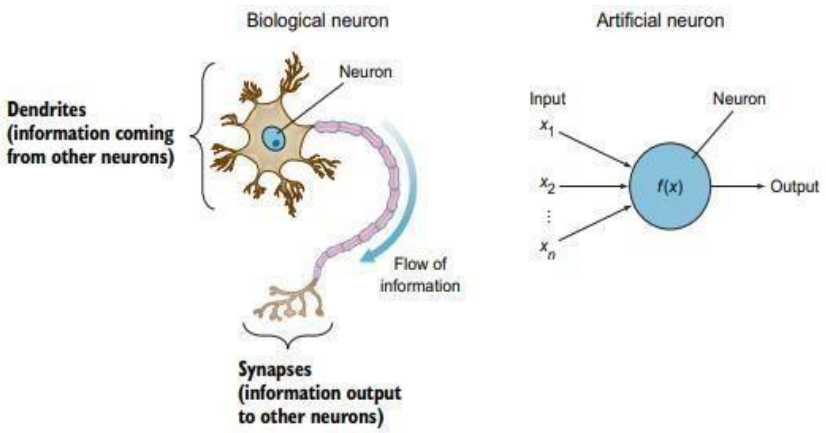
X Ray

Lidar in Drones

Interpreting devices

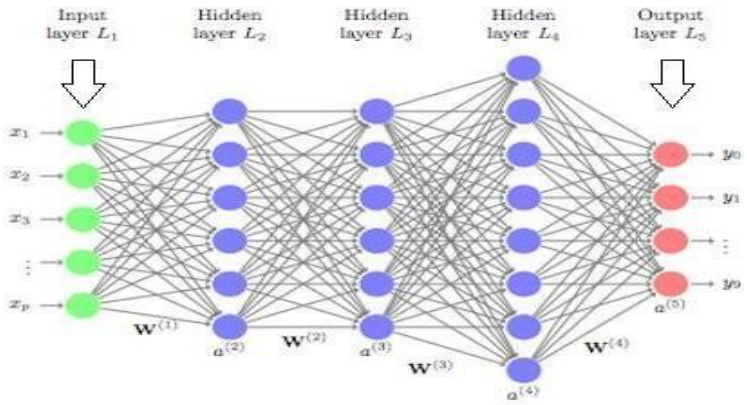
Interpreting devices in vision systems take the output image from the sensing device and learn features and patterns to identify objects. So, we need to build a brain. Thus, we have ANNs and CNNs

Artificial neural networks (ANNs) the analogy between the biological neurons and artificial systems has a main processing element, a neuron, with input signals (x1, x2,…, xn) and an output.



Similarities between biological neurons and artificial systems

Building a multilayer neural network is called Deep Learning. The DL methods learn the representations through a sequence of transformations of data through layers of neurons. The different DL architectures are ANNs and Convolutional neural networks CNNs.

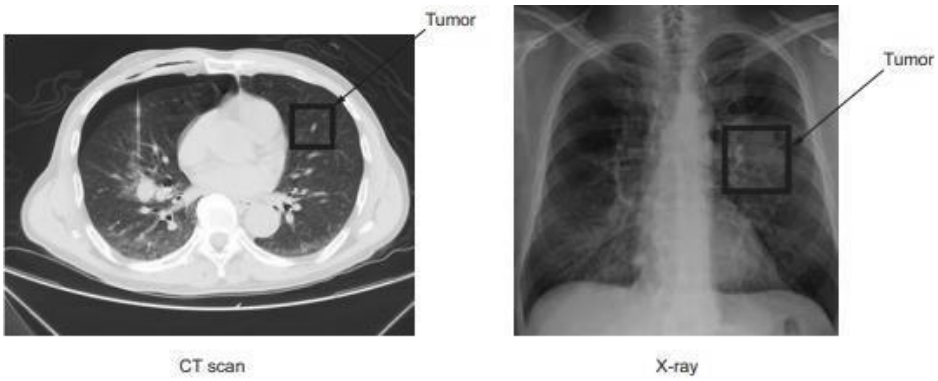


Deep learning involves layers of neurons in a network.

3. Applications of Computer Vision:

1. **Image classification**: It is the task of assigning to an image a label from a predefined set of categories.

   Example: Lung cancer diagnosis

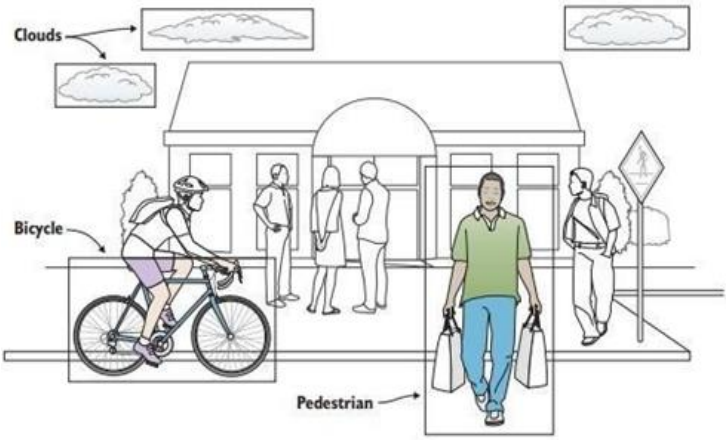


X-ray images to identify tumors in earlier stages of development.

2. **2. Traffic sign recognition:** Traditionally, standard CV methods were employed to detect and classify traffic signs, but this approach required time-consuming manual work to handcraft important features in images.

Vision systems can detect traffic signs with very high performance

3. **Object detection and localization**: Object detection goes beyond by identifying and categorizing all objects within the image. Localization involves determining the object's position and outlining it with a bounding box.



Deep learning systems can segment objects in an image.

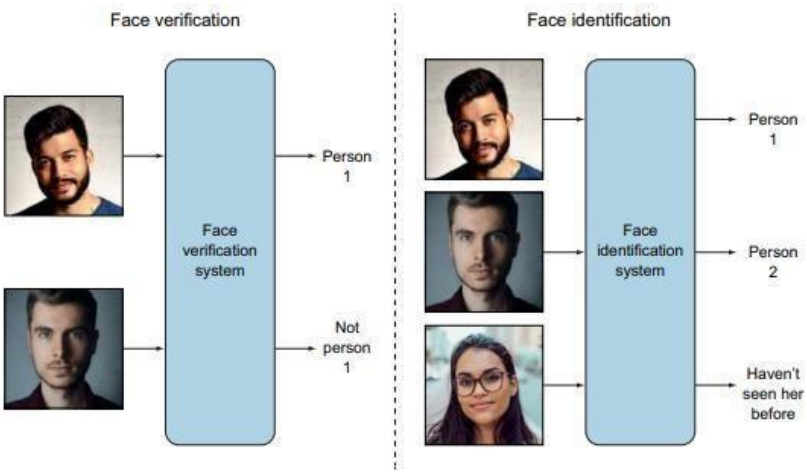
4. **Generating art (style transfer):** Neural style transfer, one of the most interesting CV applications, is used to transfer the style from one image to another.

Style transfer onto the original image, producing a piece of art that feels

5. **Creating images:** A new DL model that can imagine new things called generative adversarial networks (GANs). GANs are sophisticated DL models that generate stunningly accurate synthesized images of objects, people, and places, among other things.

Generative adversarial networks (GANS)

6. **Face recognition:** Face recognition (FR) allows us to exactly identify or tag an image of a person. Face recognition is a form of fine-grained classification. Face Recognition system is categorized in two modes:
   i. **Face identification** involves one-to-many matches that compare a query face image against all the template images in the database to determine the identity of the query face.
   ii. **Face verification** involves a one-to-one match that compares a query face image against a template face image whose identity is being claimed.



Example of face verification (left) and face recognition (right)

7. **Image recommendation system**: In this a user seeks to find similar images with respect to a given query image.

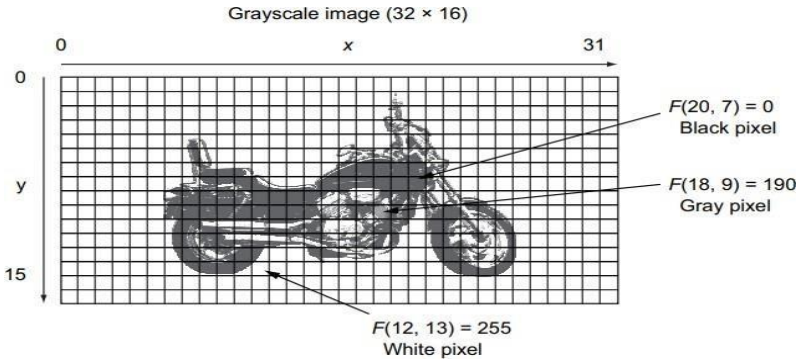
Apparel search.

8. **Input Data (Image)**: For Computer Vision applications, the input data is either images or video data.

The images are of two types, Grayscale (BW) and colour images (RGB) and videos are stacked sequential frames of images.

i. **Grayscale Image:** In gray scale images are two-dimensional and lie on the x-y plane. The gray scale image has pixel values range from 0 to 255.
Since the pixel value represents the intensity of light, the value 0 represents very dark pixels (black), 255 is very bright (white), and the values in between represent the intensity on the grayscale.

Example: Motorcycle and the grid of pixel



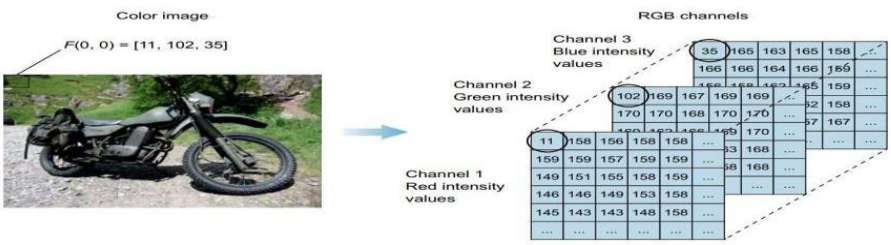
Images consists of raw building blocks called pixels.

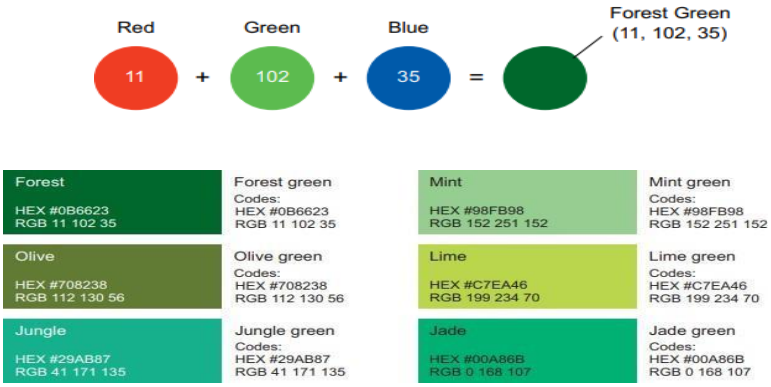
Grayscale => F(x, y) gives the intensity at position (x, y).
The pixel located at x = 12 and y = 13 is white; F (12, 13) = 255.

Mathematically an image is a function of F(x, y) and transforms it to a new image function G(x, y).

ii. **Colour images**: In colour images, the value of the pixel is represented by three numbers: the intensity of red, intensity of green, and intensity of blue.

In an RGB system, Color image in RGB => F(x, y) = [red (x, y), green (x, y), blue (x, y)]. Color image is having three matrices stacked on top of each other; that's why it's a 3D matrix.





Color images and representation

9. **Image Pre-processing**: Image data pre-processing converts image data into a form that allows machine learning algorithms to solve it. It is often used to increase a model's accuracy, as well as reduce its complexity.

The important steps in image pre-processing techniques are

i. **Grayscale conversion**: It is simply converting images from colored to black and white. It is normally used to reduce number of pixels that need to be processed and reduce computation complexity in machine learning algorithms.

Colour image transformation to gray scale

ii. **Normalization**: Image normalization is a typical process in image processing that changes the range of pixel intensity values.

Linear normalization of a digital image adjusts the pixel intensity values to a common scale, typically to improve contrast or prepare the image for further processing.

The formula for linear normalization is often expressed as:

$$I' = \frac{I - I_{\min}}{I_{\max} - I_{\min}} \times (new_{\max} - new_{\min}) + new_{\min}$$

Where:

- I is the original pixel intensity.
- $I_{\min}$ and $I_{\max}$ are the minimum and maximum pixel intensity values in the original image, respectively.
- $new_{\min}$ and $new_{\max}$ are the desired minimum and maximum intensity values in the normalized image.
- I' is the normalized pixel intensity.

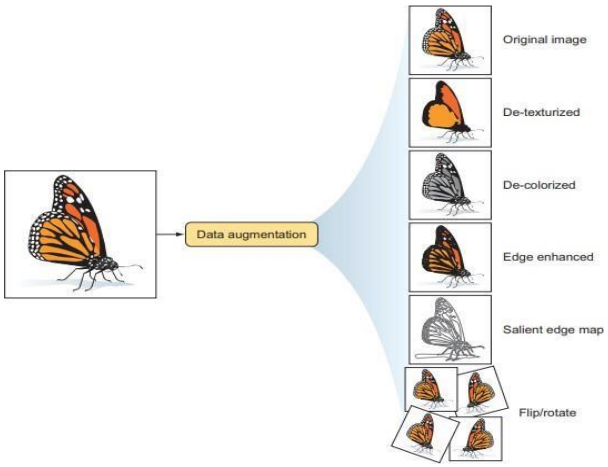
Ex: The left image depicts the right-side original image is very bright results with better contrast after the normalization process.

iii. **Data Augmentation**: Data augmentation helps in preventing a neural network from learning irrelevant features. This results in better model performance. It is the process of making minor alterations to existing data to increase its diversity without collecting new data. It is a technique used for enlarging a dataset.

There are two types of augmentation:

Offline augmentation is used for small datasets.
Online augmentation is used for large datasets. It is normally applied in real-time.

Standard data augmentation techniques include horizontal & vertical flipping, rotation, cropping, shearing, etc.



- Shifting is the process of shifting image pixels horizontally or vertically.
- Flipping reverses, the rows or columns of pixels in either vertical or horizontal cases, respectively.
- Rotation process involves rotating an image by a specified degree.
- Changing brightness is the process of increasing or decreasing image contrast.
- Cropping is the process of creating a random subset of an original image which is then resized to the size of the original image.
- Scaling image can be scaled either inward or outward. When scaling an image outward, the image becomes more significant than the original and vise versa.

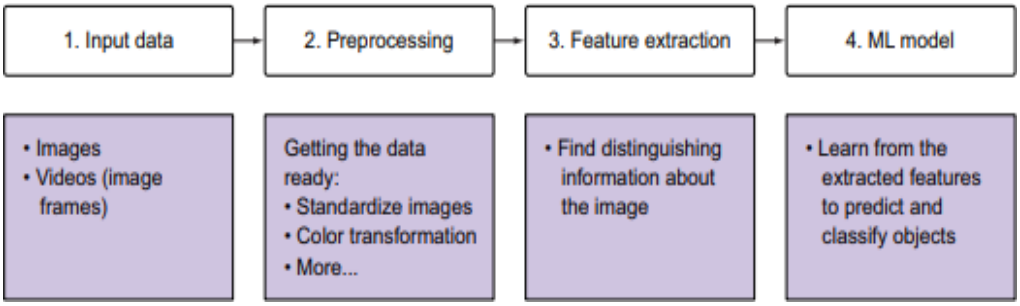
iv. **Image standardization**: Standardization is a method that scales and pre-processes images to have similar heights and widths. It re-scales data to have a standard deviation of 1 (unit variance) and a mean of 0. Standardization helps to improve the quality and consistency of data.

4. Computer Vision Pipeline Process:

The vision systems are composed of two main components,

- Sensing devices capture the image or signals through camera or sensors.
- The interpreting device is used to process and understand images.

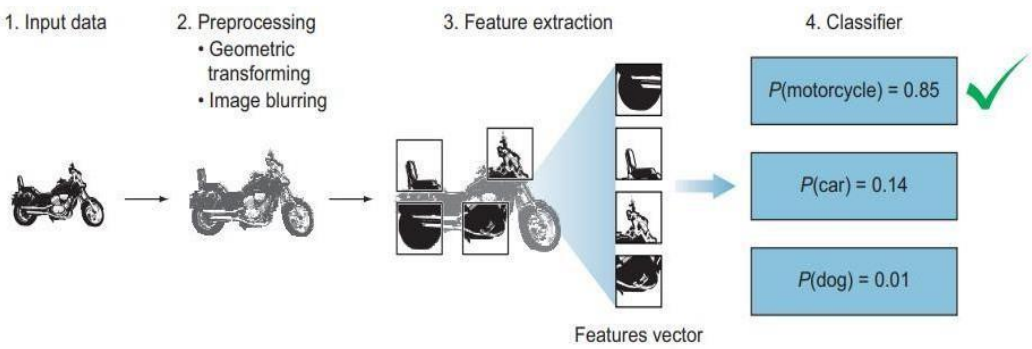
The interpreting process uses "computer vision pipeline" that consists of four steps



The computer vision pipeline

Computer Vision pipeline for image classification: An image classifier is an algorithm that takes in an image as input and predicts a label or "class" that identifies that image. A class (also called a category) in machine learning is the output category of your data.

Ex. For a motorcycle image, if we want the model to predict the probability of the object from the following classes: motorcycle, car, and dog



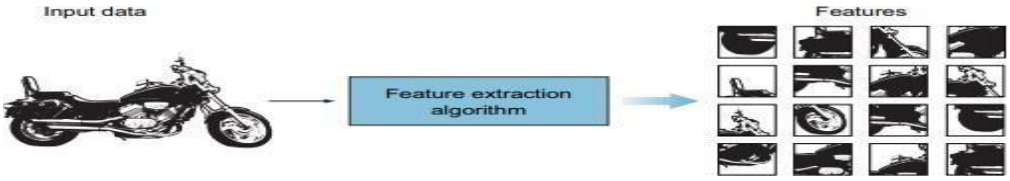
Predict the probability of the motorcycle object from the motorcycle, car, and dog classes

Feature extraction:

Features are parts or patterns of an object in an image that help to identify image. The entire DL model works around the idea of extracting useful features that clearly define the objects in the image.

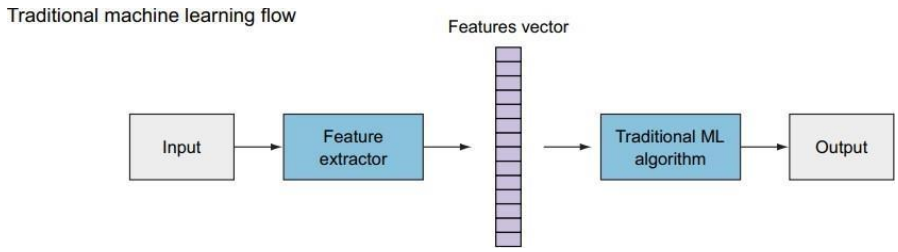
A raw data (image) is transformed into a feature vector using learning algorithm, which can learn the characteristics of the object.

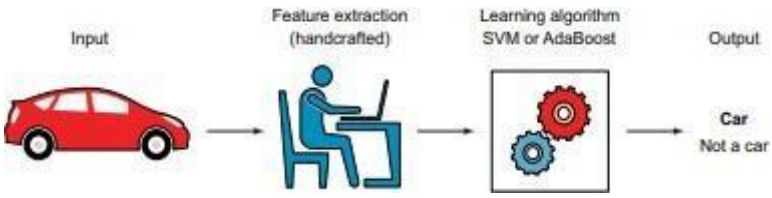
Ex: When we feed the raw input image of a motorcycle into a feature extraction algorithm. The extraction algorithm produces a vector that contains a list of features. This feature vector is a 1D array that makes a robust representation of the object.



Feature Extraction

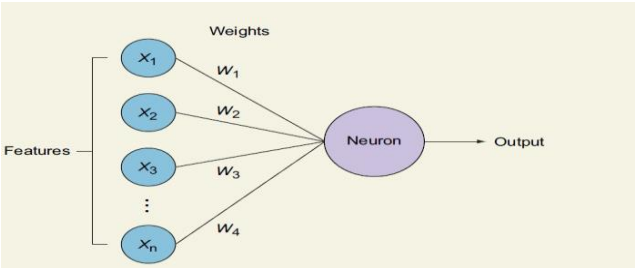
Feature Extraction for Traditional Machine Learning: The Process relies on domain knowledge (or partner with domain experts) to extract features that make ML algorithms work better. Feeding the produced features to a classifier like a support vector machine (SVM) or AdaBoost to predict the output.





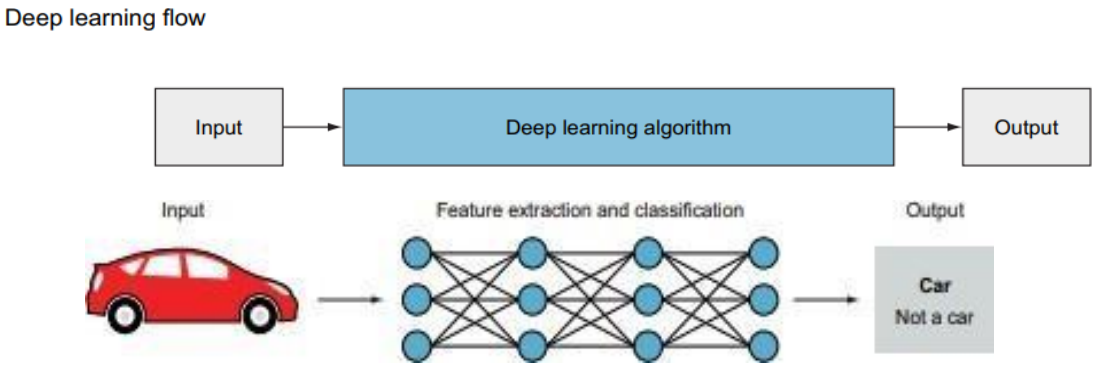
Traditional machine learning algorithms require handcrafted feature extraction.

Feature extraction in Deep Learning: In deep Learning, no need to manually extract features from the image. The network extract features automatically and learns their importance on the



output by applying weights to its connections.

When a raw image is fed to the network, while passing through the network layers, identifies features/ patterns within the image Neural networks can be thought of as feature extractors plus classifiers that are end-to-end trainable, as opposed to traditional ML models that use handcrafted features.

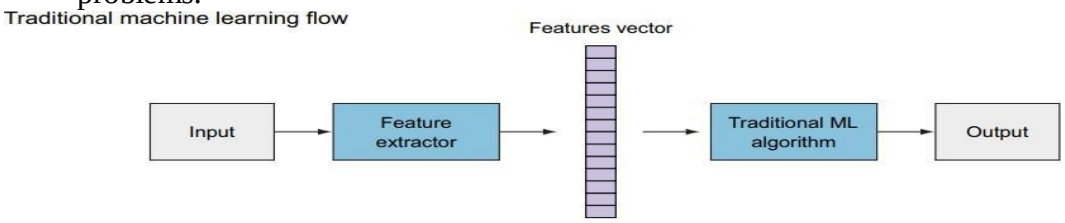


A DNN the input image through its layers to automatically extract features

5. Classifier Learning Algorithm:

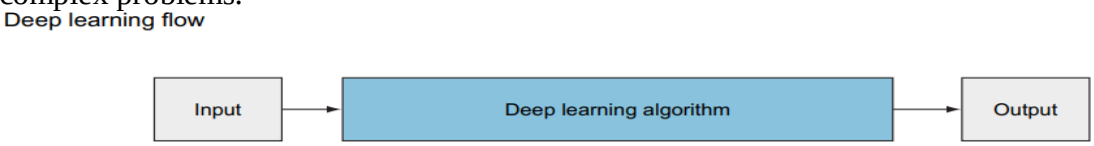
The classification task is done by two ways:

- By traditional ML algorithms like SVMs which might give result for some problems.



Traditional Classifier learning algorithm

Deep neural network algorithms like CNNs give best results to classifying images for most complex problems.



Deep NN Classifier learning algorithm

Deep Learning Classifier: The neural networks classifier automatically extracts useful features from the dataset (images), to predict the class labels for new images.

The input images are passed through the hidden layers of the deep neural network to learn their features layer by layer.

The deeper network with more hidden layers learns the features of the dataset will prone the model to overfitting and reduces the accuracy of the model.

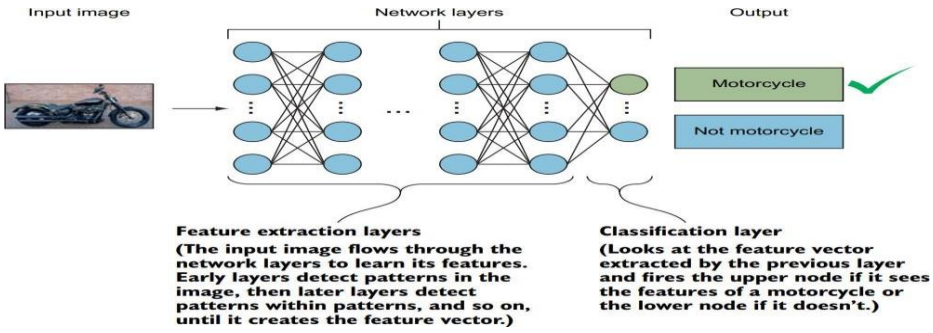
The last layer of the neural network is fully connected usually acts as the classifier that outputs the class label.

To improve the accuracy of the model, different approaches are used:

- acquire more training images, or
- more processing to remove noise, or
- extract better features, or
- change the classifier algorithm and tune some hyperparameters, or
- even allow more training time.

Deep learning Flow:

Deep NN Classifier learning algorithm



Deep NN Classifier learning algorithm

Q&A

1. A robotics company wants to build a drone that can inspect damaged buildings after an earthquake. The drone must capture surroundings and identify cracks automatically.
What two major components must the system include? Explain their roles.

Answer:
The system must include:
Sensing device – such as a camera or LiDAR to capture visual data.
Interpreting device – such as algorithms or neural networks (ANN/CNN) to analyze images and detect cracks.
The sensing device captures raw environmental data, while the interpreting device processes and understands it, similar to how the human eye and brain work.

2. An image classification system predicts whether an image contains a motorcycle, car, or dog.
What is the final output of this system?

Answer:
The system outputs a probability distribution across predefined classes (motorcycle, car, dog) and selects the class with the highest probability.

3. Before classification, the system converts the raw image into a 1D numerical representation.
What is this process called?

Answer:
Feature Extraction
It transforms the image into a feature vector, which is a 1D array representing important characteristics of the object.

4. A security system compares a person's face with all stored faces in a database to identify them.
What type of face recognition is this?

Answer:
Face Identification (One-to-Many matching)
It compares the query image with all templates in the database.

5. A person enters an ID number, and the system verifies whether the face matches the claimed identity.
What type of face recognition is this?

Answer:
Face Verification (One-to-One matching)
It compares the query face with one specific template.

6.An online fashion store allows users to upload a shirt image and find similar shirts.
Which computer vision application is used?

Answer:
Image Recommendation System
It finds similar images based on feature similarity.

7.A system outlines objects in an image with bounding boxes.
Is this image classification or object detection?

Answer:
This is Object Detection and Localization.
Because it identifies multiple objects and specifies their positions using bounding boxes.

8.An AI system transfers Van Gogh's painting style onto a photograph.
Which CV technique is being used?

Answer:
Neural Style Transfer
It transfers artistic style from one image to another.

9.A model generates completely new realistic human faces that do not exist.
Which deep learning model is responsible?
Answer:
Generative Adversarial Networks (GANs)
GANs generate synthesized images that look realistic.

10.Why can computers not "see" images the way humans do?
Answer:
Computers interpret images as numerical pixel matrices, not visual scenes.
They require algorithms to extract features and assign meaning.

11.A startup is building a system that identifies defective products from factory images.
The team has installed cameras and developed a CNN model.
Explain how this system follows the computer vision pipeline.
Answer:
The system follows these steps:

1. **Sensing Device: Cameras capture product images.**
2. **Pre-processing: Images may be resized, normalized, or converted to grayscale.**
3. **Feature Extraction: CNN layers automatically extract features such as edges, textures, and shapes.**
4. **Classification: Final fully connected layer predicts whether the product is defective or non-defective.**

12.An image classification model performs poorly due to lighting variations in training images.
Which pipeline stage should be improved and how?
Answer:
The Pre-processing stage should be improved.

Possible solutions:

- **Apply normalization**
- **Adjust brightness**
- **Perform grayscale conversion**
- **Apply noise removal**

This ensures consistent input for feature extraction.